RENEWABLE ENERGY*

Renewable Energy is type of energy that contains naturally unlimited supply and restore again over time. For example solar energy, wind energy, hydropower, biomass, geothermal energy.

SOLAR ENERGY*

Solar energy is the energy of the sun. It can be utilized through Photovoltaic (PV) Cells and Solar Thermal Collectors, (How Solar Energy Works)

PHOTOVOLTAIC CELLS (PV) / SOLAR PANELS*

These are devices that convert sunlight directly into electricity. They are made up of a series of interconnected solar cells, each of which is composed of two layers of semiconductor material, typically silicon. When sunlight strikes a solar cell, it energizes the electrons in the semiconductor, causing them to flow and generate electricity.

Types of Solar Panels/PV Cells

- **Monocrystalline Solar Panels: (20%)**
  These panels are made from single crystal of silicon. The purest and most efficient type of solar cell but most expensive also. Black in color, have a shiny finish

- **Polycrystalline Solar Panels:** (15%)
  Made-up from multiple crystals of silicon, which are less pure and less expensive than monocrystalline silicon. Blue or gray in color and have a matte finish.

- **Amorphous/Thin Film Solar Panels (8%) ***
  Made-up from non-crystalline silicone that can be deposited onto a variety of substrates, including glass, plastic. Least efficient but they are the cheapest option. They work well in low light, even moonlight. Following are its examples:

1. **Amorphous Silicon (a-Si):** (8%)
   a non-crystalline form of silicon that has a random, disordered atomic structure. This makes it a less efficient conductor but also makes it easier to manufacture and cheaper.

2. **Cadmium Telluride (CdTe):** (15%)
   This material offers good efficiency.. However, it contains cadmium, a toxic heavy metal, which raises environmental problems and causes respiratory issues.

Benefits of PV Cells (Solar Energy)

- Clean: Doesn't produce any greenhouse gases
- Sustainable: Doesn't require any fossil fuels to operate.
- Quiet: Doesn't create any noise pollution.

Applications of PV Cells

- Electricity generation for homes and businesses.
- Providing backup power in case of a power outage.
- Powering electric vehicles to move.

SOLAR THERMAL COLLECTORS

It is a type of heat exchanger that converts solar radiation into thermal energy.
PTC is the example that provides 60-70% efficiency. It uses the mirror surface to focus direct solar radiation on the receiver tube that contains a working fluid. This fluid absorb the concentrated solar energy then heat exchanger transfers the heat from the HTF to another fluid, such as water for use in applications residentially or commercially

STORAGE OF RENEWABLE ENERGY*

Batteries are the most common way to store solar energy. They work by converting the direct current (DC) electricity generated by solar panels into stored chemical potential energy. When the battery is charged, an electric current is used to move lithium ions from the anode to the cathode creating a chemical potential difference between two electrodes. When the battery is discharged, the lithium ions move back from the cathode to the anode, releasing electrical energy in the process.

Hydroelectric Energy Storage: This method stores energy by pumping water from a lower reservoir to a higher reservoir. When energy is needed, the water is released from the higher reservoir, generating electricity as it flows through turbines. Efficient technology with large storage capacity.

WIND ENERGY*

Wind power utilizes the K.E of wind to generate electricity. As the wind blows, the blades rotate, turning a shaft that is connected to a generator. The generator converts the M.E of the rotating shaft into electrical energy. This electricity can then be used to power homes, businesses, or the grid.

WIND TURBINE COMPONENTS

Tower: The tower supports the wind turbine and its components.

1. Rotor: It is the rotating part of the turbine; it consists of blades and a hub, central part connecting the blades. Hub holds the blades and connects them to the main shaft

2. Generator: The generator converts the M.E of the rotating shaft into E.E.

3. Yaw system: It is used to turn the wind turbine rotor against the wind.

4. Pitch system: The pitch system controls the angle of the blades.

TYPES OF WIND TURBINES

Horizontal-Axis Wind Turbines (HAWTs)	Vertical-Axis Wind Turbines (HAWTs)
High Efficiency 45% Parallel to the direction of the wind HAWTs require alignment with the wind direction, needs a yaw system	Lower Efficiency 30% Perpendicular to the direction of wind VAWTs can capture wind energy from any direction, eliminating the need of yaw system

TURBINES

Turbines are rotating machines that convert the energy of a fluid, such as air, water, or steam, into mechanical energy. Applications, including power generation, propulsion, and industrial processes.

Steam Turbines:
Widely used in fossil fuels based power plants to generate electricity. They operate on the principle of expanding steam through a series of blades, converting the thermal energy of the steam into mechanical energy.

Hydraulic Turbines:
They are used in hydroelectric power plants to convert the kinetic energy of flowing water into mechanical energy.

Wind Turbines:
They harness the kinetic energy of wind to generate electricity.

HYDROPOWER ENERGY

Hydroelectricity harnesses the energy of moving water to generate electricity. In hydroelectric power plants, water from a reservoir is directed through a turbine, which converts the kinetic energy of the flowing water into mechanical energy. This mechanical energy is then converted into electricity through generator.

TIDAL ENERGY*

Tidal energy is a form of renewable energy that uses the power of the tides to generate electricity. Tides are the rise and fall of sea levels caused by the gravitational pull of the moon and sun. As the water moves in and out, it creates a flow of energy that can be captured and converted into electricity.

Tidal Stream Generators; They are similar to underwater wind turbines. They consist of blades that are placed in areas with strong tidal currents. As the water flows past the blades, it turns them, generating electricity by converting the K.E of water into M.E and then generators convert the mechanical energy into electrical energy.

Pros and Cons of Tidal Energy

- Tidal energy is a very abundant resource.
- Clean form of energy that does not produce greenhouse gas emissions
- High Initial Costs

BIOMASS ENERGY

Biomass is organic matter from plants or animals. It can be burned directly to produce heat or converted into biofuels, such as ethanol or biodiesel, which can be used to power vehicles.

GEOTHERMAL ENERGY

Geothermal energy is the heat from the Earth's interior. It can be used using geothermal power plants, which extract heat from the ground to generate electricity or heat buildings.

SOURCES OF RENEWABLE ENERGY*

- Sunlight is the most abundant source of renewable energy on Earth, regardless of our location. (Solar energy through PV cells)
- Wind is another source of renewable energy (Wind energy through Wind Turbines)

- Water is a renewable resource that has been used for centuries (Hydropower energy by Reservoir and Turbines)
- Plants and animals are a source of biomass. Biomass can be burned directly to produce heat or converted into biofuels, which can be used to power vehicles

	Advantages	Disadvantages
Solar	Pollution-free and reduced electricity bill	No solar power on cloudy days and night
Wind	Wind energy – low operating expenses	Wind turbines – expensive to install Wind blades – safety concern
Biomass	Clean sources of energy and biomass energy could be stored	Required more land and sustainable management

THE GREENHOUSE EFFECT *

A natural process that helps to keep Earth's temperature at a livable level. It occurs when gases in the atmosphere trap some of the sun's energy, preventing it from escaping back into space. This trapped energy warms the planet.
It includes carbon dioxide, methane, nitrous oxide, and ozone. These gases are essential for life on Earth, but human activities are increasing their concentrations in the atmosphere which is very dangerous causing extreme weathers

Sources/Reasons of Greenhouse Gases:

Deforestation – Cutting of trees so they can't absorb CO_2 now
Vehicle Emissions – Produces CO_2
Industrialization – Burning of Fossil fuels like coal

Side effects:

Global warming - The rise in the average temperature of Earth's atmosphere
Depletion of ozone layer – Protection layer around earth that protect us from U.V rays coming from the sun. U.V can cause skin problems.
Rising sea levels: As the temperature of the oceans rises, they expand and glaciers and ice sheets melt, causing sea levels to rise. This threatens coastal communities

Solutions:

Transition to renewable energy sources
Reduce deforestation

BATTERIES *

Batteries are devices that store chemical energy and convert it into electrical energy. Applications: Powering our smartphones and laptops, providing backup power for homes and businesses.

Types of Batteries:

- **Primary Batteries**: These batteries cannot be recharged because the chemical reactions that occur within them are irreversible.
  Dry cell (Electrolyte-Ammonium chloride and Zinc chloride paste) – Flashlights and Remotes
  Mercury button cell – Watches and Calculators

- **Secondary Batteries** These batteries can be recharged because the chemical reactions that occur within them are reversible.
  Lead storage battery (Electrodes-Lead , Electrolyte-H_2SO_4) – Automobiles
  Lithium-ion battery – Mobile batteries

Applications of Batteries

- Providing Power to Portable Electronic Devices

- Powering Automobiles

- Provide backup power in case of a power outage

Electrolysis Process *

Electrolysis is an electrochemical process that uses an electric current to drive non-spontaneous chemical reactions. It involves using an electrolyte; a solution that conducts electricity, and two electrodes (Anode and Cathode) connected to the terminals. When an electric current is applied, the electrolyte ions move toward the electrodes, and chemical reactions occur at each electrode.

Electrolysis can be used to produce Hydrogen gas

- Hydrogen gas (H_2): This is produced when water (H_2O) is electrolyzed, with the hydrogen gas produced at the cathode and oxygen gas produced at the anode.

Chemical Energy into Electrical Energy

Batteries and fuel cells are the example of process using chemical reactions to generate electricity.

Common electrolytes used in electrolysis include:

- Water (H2O): Used to produce hydrogen and oxygen gas.
- Brine: Solution of Sodium Chloride (NaCl) in water. Use to produce chlorine and sodium hydroxide (NaOH).

Catalysts *

Substances that alter the rate of a chemical reaction without being consumed in the reaction itself.

1. **Homogeneous Catalysts:**
   - **Definition:** Homogeneous catalysts are in the same phase (usually liquid or gas) as the reactants.
   - **Example:** (H_2SO_4)
2. **Heterogeneous Catalysts:**
   - **Definition:** Heterogeneous catalysts are in a different phase (usually solid) than the reactants.
   - **Example:** Platinum, Nickel
3. **Photo catalyst, Titanium dioxide (TiO2)**
   - Changes the rate of a chemical reaction on exposure to light. TiO2 absorbs photons in the presence of sunlight, generating electron-hole pairs that can participate in chemical reactions. TiO2 is a non-toxic and abundant material, making it an environmentally friendly
   - Photo electrochemical Water Splitting: TiO2 can be used to split water molecules into hydrogen and oxygen in the presence of sunlight.

Positive catalysts increase the rate of a reaction by lowering the activation energy. The activation energy is the minimum amount of energy that the reactants must have in order for the reaction to occur. By lowering the activation energy, the catalyst makes it more likely for the reactants to collide with enough energy to form products
Sulphuric acid, MnO2 or V2O5

Negative catalysts, also known as inhibitors, slow down the rate of a reaction by increasing the activation energy. They do this by interfering with the collision of the

reactants or by stabilizing the reactants in their original form.
Alcohol

- iii. Promoter. It is added to a **catalyst** to increase its efficiency. **Example**: Molybdenum.

PARTICLE AS CATALYST *

Yes, numerous particles can be used as catalysts,

1. Metals: Metal particles, such as platinum, gold, are among the most widely used catalysts due to their high reactivity. They are often used in hydrogenation reactions, oxidation reactions, and polymerization reactions.

Reaction without a catalyst:

- Decomposition of hydrogen peroxide:

  Hydrogen peroxide (H_2O_2) decomposes into water and oxygen over time:

  $2H_2O_2 \rightarrow 2H_2O + O_2$

  This reaction is very slow at room temperature. If you leave a bottle of hydrogen peroxide open, you will notice that it slowly bubbles as oxygen is released.

Reaction with a catalyst *

- Decomposition of hydrogen peroxide with manganese dioxide:

  Manganese dioxide (MnO_2) is a catalyst that speeds up the decomposition of hydrogen peroxide:

  $2H_2O_2 + MnO_2 \rightarrow 2H_2O + O_2 + MnO_2$

  This reaction is much faster than the reaction without a catalyst. If you add a small amount of manganese dioxide to a bottle of hydrogen peroxide, you will see it bubble vigorously as oxygen is released.

FUEL CELL *

A fuel cell is an electrochemical device that converts the chemical energy of a fuel into electrical energy. It does this by combining a fuel, such as hydrogen or methane, with an oxidizer, such as oxygen, to produce electricity, water, and heat.

Types of Fuel Cells

There are many types of fuel cells:

1. **Proton Exchange Membrane (PEM) Fuel Cells:**

Electrolyte: Proton-conductive membrane typically consists of a polymer.
Fuel: Hydrogen
Applications: Suitable for Transportation, Backup Power

2. **Alkaline Fuel Cells (AFC) or Bacon Fuel Cell:**

Electrolyte: Alkaline electrolyte, potassium hydroxide (KOH).
Fuel: Can use a variety of fuels, including hydrogen, methanol, and ammonia.
Applications: Historically used in space missions and produced potable water, low-grade heat, and electricity

Benefits of Fuel Cells

Fuel cells offer several advantages over traditional fossil fuel-powered generators:

Fuel cells produce only water as by products, making them a clean energy
Fuel cells possess 60% whereas traditional combustion engines possess 30%
Fuel cells can be used to power electric vehicles
Fuel cells are used as power backup for commercial and residential buildings and in remote areas.

Challenges of Fuel Cells

- Fuel cells are currently expensive
- More complex to manufacture and operate
- Safety: Fuel cells can be flammable

HYDROGEN FUEL CELL *

A hydrogen fuel cell uses the chemical energy of hydrogen to produce electricity. It is a clean form of energy with **electricity**, heat and water being produced,

HOW DOES A FUEL CELL WORK *

A hydrogen fuel cell consists of two electrodes (anode and cathode) separated by an electrolyte. The anode is where the hydrogen fuel enters the cell, and the cathode is where the oxygen enters the cell. The electrolyte is a material that allows ions to move between the electrodes.

When hydrogen gas (H2) enters the anode, it is split into protons (H+) and electrons (e-). The protons pass through the electrolyte and reach the cathode. The electrons flow through an external circuit, creating an electrical current. At the cathode, the protons combine with oxygen (O2) and electrons to form water (H2O).

The overall reaction in a hydrogen fuel cell is:

2H2 + O2 → 2H2O + energy

HYDROGEN AS FUEL AND METHODS OF GENERATING H2 GAS *

Hydrogen (H2) is a versatile fuel with the potential to power a variety of applications, including transportation, power generation, and industrial processes. However, one of the key challenges in utilizing hydrogen as a fuel is its production. Hydrogen is not found in its pure form on Earth but different methods of generating H2 gas:

1. Steam Methane Reforming (SMR): Methane (natural gas) is reacted with steam over a nickel catalyst at high temperatures to produce hydrogen and carbon monoxide.

$$CH_4(g) + H_2O(g) \longrightarrow 3H_2(g) + CO(g)$$

$$\Delta H^{\ominus} = +210 \text{ kJ mol}^{-1}$$

2. Electrolysis: Electrolysis is a process that uses electricity to split water (H2O) into hydrogen and oxygen. This method produces pure hydrogen.

Converting platinum catalyst into nano-size particles *

Chemical Reduction: This method involves reducing platinum ions to metallic platinum nanoparticles using reducing agents like sodium citrate (C6H5Na3O7). The size and

shape of the nanoparticles can be controlled by adjusting the reaction conditions like concentration of the reducing agent, temperature, and reaction time.

Cutting a 100nm thin layer of a material *

Focused Ion Beam (FIB) Milling: It is a process that utilizes a focused beam of energetic ions to bombard and remove away material from a surface. This process relies on sputtering, in which energized ions physically eject other atoms and molecules from the sample surface

DOPING *

Doping is a technique used to intentionally introduce impurities into a pure semiconductor to modify its electrical properties. Normally Trivalent and Pentavalent elements are used to dope Silicon and Germanium semiconductors. Doping plays a crucial role in the fabrication of various semiconductor devices, including **solar cells**, transistors, and light-emitting diodes (LEDs).

N-type semiconductor	P-type semiconductor
1. It is an extrinsic semi-conductor which is obtained by doping the impurity pentavalent impurity atoms such as antimony, phosphorous, arsenic etc. to the pure germanium or silicon semiconductor.	1. It is an extrinsic semiconductor which is obtained by doping trivalent impurity atoms such as boron, gallium, indium etc. to the pure germanium or silicon semiconductor.
2. The impurity atoms added, provide extra electrons in the structure, and are called donor atoms.	2. The impurity atoms added, create vacancies of electrons (i.e., holes) in the structure and are called acceptor atoms.
3. The electrons are majority charge carriers and holes are minority charge carriers.	3. The holes are majority charge carriers and eletrons are minority carriers.

JUNCTION IN A SOLAR CELL *

A junction in solar cells is a critical component that plays a vital role in converting sunlight into electricity. It is formed by joining two different types of semiconductor materials, typically p-type and n-type semiconductor. When sunlight hits the junction, it energizes the electrons in the n-type layer. These energized electrons then cross the junction and enter the p-type layer. This flow of electrons creates an electric current,

• Renewable energy is derived from natural sources that are constantly replenished, such as sunlight, wind, water. • Sustainable source because they are infinite. • Cleaner, producing less greenhouse gas emissions and pollution.	• Fossil fuels are formed from the remains of ancient plants and animals. These include coal, oil, and natural gas. • Fossil fuels are finite resources • Releases significant amounts of greenhouse gases and pollution.

PETROL ENGINE VS DIESEL ENGINE

Petrol engines and diesel engines are both internal combustion engines that convert chemical energy from fuel into mechanical energy to power vehicles.

Combustion Process:
Petrol engines use spark ignition, where a spark plug ignites the air-fuel mixture in each cylinder. Diesel engines use compression ignition, where the air-fuel mixture is compressed to a high temperature that causes it to spontaneously ignite.

Power and Torque:
Diesel engines typically produce more torque that accelerates a vehicle, at lower RPMs. Suitable for heavy-duty applications like trucks and buses. Petrol engines produce more power at higher RPMs. Suitable for lighter vehicles like bikes and cars

Efficiency:
Diesel engines are generally more fuel-efficient than petrol engines because they have a higher compression ratio

THERMODYNAMIC CYCLES

In mechanical engineering, various thermodynamic cycles are employed in power plants to convert heat into electricity. Here are some of the most common thermodynamic cycles used in power plants:

1. Rankine Cycle:

The Rankine cycle is the most widely used thermodynamic cycle for steam power plants. Working fluid-Water. It involves four main processes:

- Evaporation: Water is heated in a boiler to produce steam.
- Expansion: The steam is expanded through a turbine, converting its thermal energy into mechanical energy.
- Condensation: The exhaust steam from the turbine is condensed back into water in a condenser.
- Pumping: The condensed water is pumped back to the boiler, completing the cycle.

2. Brayton Cycle:
   A thermodynamic cycle that describes how gas turbines operate. Application; aircraft engines
3. Otto Cycle:
   Theoretical cycle Petrol engines
4. Diesel Cycle:
   Theoretical cycle Diesel engines
5. Combined Cycle:
   Combination of Rankine and Brayton cycles to achieve higher overall efficiency. The exhaust heat from a gas turbine is used to generate steam in a heat recovery steam generator (HRSG), which then drives a steam turbine.

PUMPS

A pump is a device that moves fluids by mechanical action, typically converted from electrical energy into hydraulic energy.

Positive displacement pumps displace the same amount of liquid per cycle.
Best selection for pumping higher-viscosity liquids such as oil, paints.
Applications: Hydraulic systems

Centrifugal pump moves a fluid by means of the transfer of rotational energy through impellers. Main parts of the pump:
• The impeller is the rotating part that converts prime mover M.E into the K.E of the fluid.
• The diffuser is the stationary part that converts the K.E of fluid into P.E.

Best selection for pumping low-viscosity liquids such water.
Application: Water supply and irrigation

HEAT AND MASS TRANSFER CONCEPTS *

- **Conduction**: Transfer of heat energy between molecules or atoms that are in direct contact with each other. The rate of heat transfer by conduction is directly proportional to the temperature difference between the two objects and the thermal conductivity of the material.
  Example: If you hold a metal spoon in a hot cup of coffee, the heat from the coffee will transfer to the spoon through conduction and this will transfer heat to your hand.

- **Convection**: Transfer of heat energy by the movement of fluids, such as liquids and gases. It occurs when there is a difference in temperature between different parts of a fluid. The rate of heat transfer by convection is directly proportional to the temperature difference between the fluid and the surrounding environment and fluid density and viscosity.
  Example: When you heat water on a stove, the water at the bottom absorbs heat from the stove so gradually whole water becomes warmer.

- **Radiation**: Transfer of heat energy through electromagnetic waves. It occurs in a vacuum, without a medium for heat transfer.
  Example: The sun's heat reaches the Earth through radiation. The sun's rays travel through the vacuum of space and are absorbed by the Earth's atmosphere and surface.

- Specific heat capacity of a substance is the amount of heat energy that must be added to 1 kg of substance to raise its temperature by 1°C

- Latent heat: The latent heat of a substance is the amount of heat energy that must be added or removed to change the substance's phase, such as from solid to liquid or from liquid to gas.

- Thermal conductivity of a material is its ability to conduct heat.

- Thermal resistance of a material is its resistance to heat flow.

- Heat flux: The heat flux is the rate of heat transfer per unit area. Unit: watts per square meter (W/m^2).

- Heat transfer coefficient: Measure of the effectiveness of heat transfer between a surface and a fluid. Unit: watts per square meter per degree Celsius ($W/m^2 \cdot K$).